

Inference-Time Scaling in Latent Space:
Recurrent Non-Verbal Reasoning

Track Submission: DataForge x Pathway — Explain the Frontier
Interactive Web Explainer: https://theshrihari.github.io/Dataforge-Submission/
Repository: https://github.com/TheShriHari/Dataforge-Submission
Topic: Test-Time Compute Scaling & Recurrent Latent Reasoning (BDH / BDH-CQ)

1. Central Claim

"Increasing the number of recurrent latent computation steps at inference time can improve reasoning accuracy without producing any additional natural-language reasoning tokens — but gains diminish and each extra step costs more compute/time."

2. Design Motivation: Latent Iteration vs. Verbalized Chain-of-Thought

Current LLM test-time scaling primarily relies on autoregressive Chain-of-Thought (CoT) prompting or search sampling (Snell et al., 2024, arXiv:2408.03314). While effective, verbalized CoT suffers from severe architectural bottlenecks:

- Context & Memory Bloat: Every verbalized reasoning token appends to the context window, causing O(N^2) self-attention cost and quadratic Key-Value (KV) cache expansion.
- Syntactic Fragility & Drift: Verbal reasoning demands strict grammatical adherence; an early hallucination or syntactic failure permanently derails subsequent steps.
- Contrast with Standard Forward Passes: Standard Transformer or feedforward CNN passes apply a static layer sequence where compute is strictly bound to parameter depth L. In contrast, recurrent latent iteration loops hidden states through a weight-shared core for K discrete iterations: h_k = Core(h_{k-1}, x) for k in [1, K] This decouples inference compute budget from parameter count, preserving a constant tensor memory footprint (h in R^{C x H x W}) with zero emitted text tokens.

3. Technical Mechanism & Measured Trade-Offs

Our implementation evaluates this mechanism on 2D maze pathfinding, where long-range connectivity requires deep propagation. Using a 123,713-parameter shared Convolutional GRU (ConvGRU) core (ml/model1.py), each recurrence step k expands the effective receptive field by 2 grid cells.

Measured across N=100 held-out 15 x 15 synthetic mazes (Seed 999, results/accuracy_by_effort.json):

- Steep Early Accuracy Phase (K = 1 -> 10): Exact solve rate climbs from 1.0% (K = 1, 0.666 ms) to 44.0% (K = 5, 2.911 ms) and 81.0% at K = 10 (6.155 ms). Path IoU rises from 55.94% to 96.61%, reflecting rapid topological discovery.
- Empirical Diminishing Returns (K = 10 -> 20): Increasing K from 10 to 20 yields only a +7.0% gain in solve rate (81.0% -> 88.0%) and plateaued Path IoU (96.61% -> 98.13%), while latency more than doubles from 6.155 ms to 12.884 ms/maze.
- Training Horizon & Generalization: K = 16-20 extends beyond the K ~ Uniform(1, 15) training curriculum (ml/train.py). The model generalizes because the recurrent core is weight-shared and lacks step-count parameters, but this was not part of the training distribution.

4. Multi-Dimensional Paradigm Comparison

Table with 4 columns: Dimension, Toy Latent Maze Solver (Our Artifact), BDH-CQ Foundation Model (Kosowski et al., 2026), and Autoregressive Search / CoT (Snell et al., 2024). Rows include Domain & Scale, Effort vs. Accuracy, Inference Hardware Cost, Memory Footprint, and Observability.

5. Architectural Roles: BDH vs. BDH-CQ

- To preserve technical precision, the two literature systems are distinguished:
- The Dragon Hatchling (BDH) (Kosowski et al., 2025, arXiv:2509.26507): A biologically-inspired, post-transformer foundational architecture that establishes recurrent latent state updates and non-autoregressive parallel processing across continuous representations.
- BDH-CQ (Kosowski et al., 2026, arXiv:2608.09888): A specialized system built within the BDH family incorporating in-context demonstration learning ("Context Queries") and test-time effort modulation across LOW, MEDIUM, and HIGH budgets without emitting scratchpad text tokens.

6. Evidence Classification: Strengths, Weaknesses, and Maturity

- Advantage Over Incumbents: Recurrent latent iteration eliminates O(N^2) KV-cache bloat and syntax drift, enabling variable-effort inference within a fixed memory envelope.
- Where It Performs Worse / Remains Untested: Untested on open-ended natural language generation where sequential text tokens are the required output modality.
- Our Experimental Results: Classified as own toy implementation, empirically measured, not independently audited on synthetic 2D mazes.
- BDH-CQ External Baseline: ARC-AGI-1 score (29.5% pass@2) is developer-reported, partially independently audited by Bielik/NYU for the headline score only. The internal scaling ablation (Table 5: 21.0% / 27.0% / 29.5% across effort tiers) is developer-reported and not independently reproduced.

7. Primary Limitation & Open Question

While recurrent latent scaling avoids token-generation overhead, it introduces a key bottleneck: latent reasoning lacks explicit symbolic auditability. If a model fails at step K = 4, diagnosing whether failure stemmed from representation saturation, gradient vanishing, or feature drift requires probing classifiers rather than reading an English scratchpad. Developing standardized, zero-overhead probing techniques for continuous recurrent latent trajectories remains a crucial open research question.

8. Where to Continue Learning

- BDH Foundations: Kosowski et al. (2025), "The Dragon Hatchling", arXiv:2509.26507 and code: github.com/pathwaycom/bdh.
- BDH-CQ Technical Report: Kosowski et al. (2026), arXiv:2608.09888.
- Test-Time Compute Theory: Snell et al. (2024), arXiv:2408.03314.
- Interactive Explainer & Verification: Live app, empirical plots, and step traces: https://theshrihari.github.io/Dataforge-Submission/.